MLflow

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LLMs & Agents

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MLflow

- MLflow is the largest open source **AI engineering platform for agents, LLMs, and ML models**.
- MLflow's tools for traditional machine learning and deep learning: ML experiment tracking, model versioning, model deployment, and model evaluation.
- **Core Feature:**
  - How to log parameters, metrics, and a model using the MLflow logging API
  - How to navigate to a model in the MLflow UI
  - How to load a logged model for inference
- **Documentation:** https://mlflow.org/docs/latest/ml/

Step 1 - Set up MLflow

- > pip install mlflow

```
import mlflow

mlflow.set_experiment("MLflow Quickstart")
```

Step 2 - Prepare training data

```
import pandas as pd
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Load the Iris dataset
X, y = datasets.load_iris(return_X_y=True)

# Split the data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Define the model hyperparameters
params = {
    "solver": "lbfgs",
    "max_iter": 1000,
    "random_state": 8888,
}
```

Step 3 - Train a model with MLflow Autologging

- In this step, we train the model and log the model and its metadata to MLflow.
- The easiest way to do this is to using MLflow's Autologging feature.

```
# Enable autologging for scikit-learn
mlflow.sklearn.autolog()

# Just train the model normally
lr = LogisticRegression(**params)
lr.fit(X_train, y_train)
```

- Now, you can focus on training the model, and MLflow will take care of the rest:
  - Saving the trained model.
  - Recording the model's performance metrics during training, such as accuracy, precision, AUC curve.
  - Logging hyperparameter values used to train the model.
  - Track metadata such as input data format, user, timestamp, etc.

Automatic Logging with MLflow Tracking

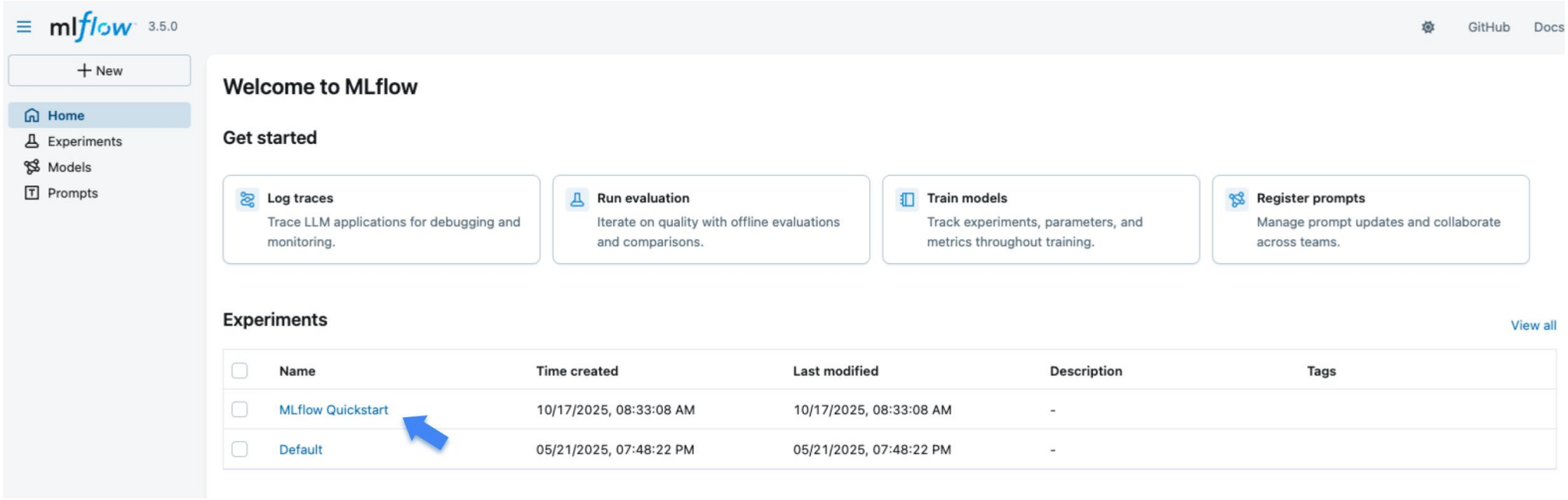
- Auto logging is a powerful feature that allows you to log metrics, parameters, and models without the need for explicit log statements. All you need to do is to call mlflow.autolog() before your training code.

```
import mlflow

mlflow.autolog()

with mlflow.start_run():
    # your training code goes here
    ...
```

- **Metrics** - MLflow pre-selects a set of metrics to log, based on what model and library you use
- **Parameters** - hyper params specified for the training, plus default values provided by the library if not explicitly set
- **Model Signature** - logs Model signature instance, which describes input and output schema of the model
- **Artifacts** - e.g. model checkpoints
- **Dataset** - dataset object used for training (if applicable), such as *tensorflow.data.Dataset*

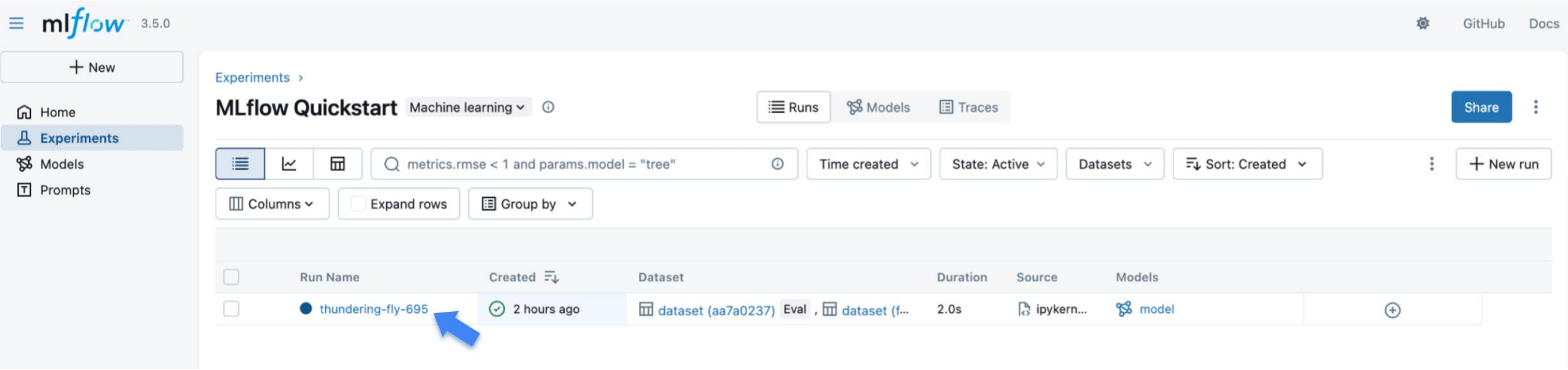
Step 4 - View the Run in the MLflow UI

- In Terminal: > mlflow ui OR mlflow server --port 5000
- Then, go to: http://127.0.0.1:5000/
- Open a new terminal, and run the code file.
- When opening the site, you will see a screen similar to the following:
- The "Experiments" section shows a list of (recently created) experiments. Click on the "MLflow Quickstart" experiment.



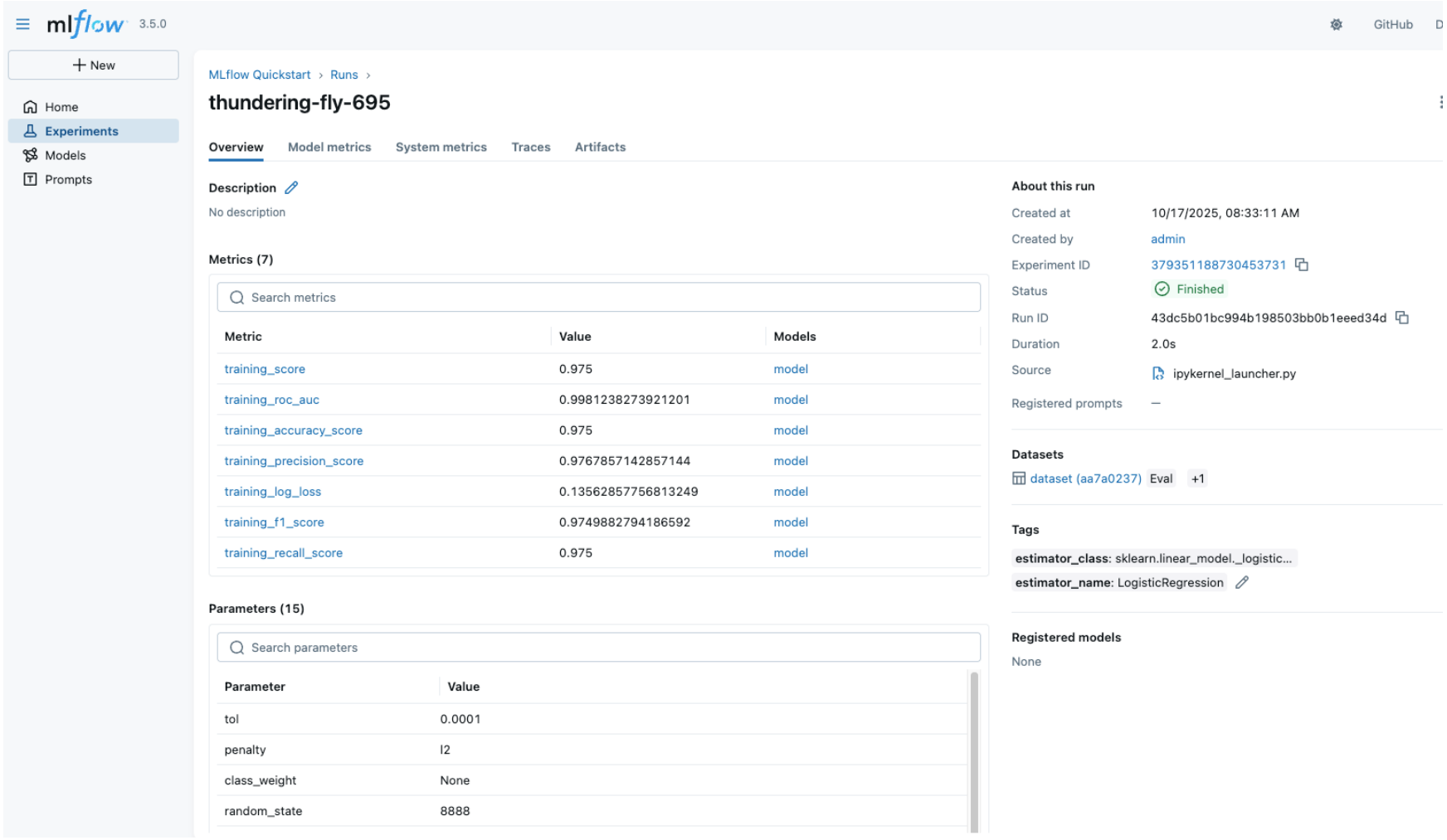
Step 4 - View the Run in the MLflow UI

- The training Run created by MLflow is listed in the table. Click the run to view the details.



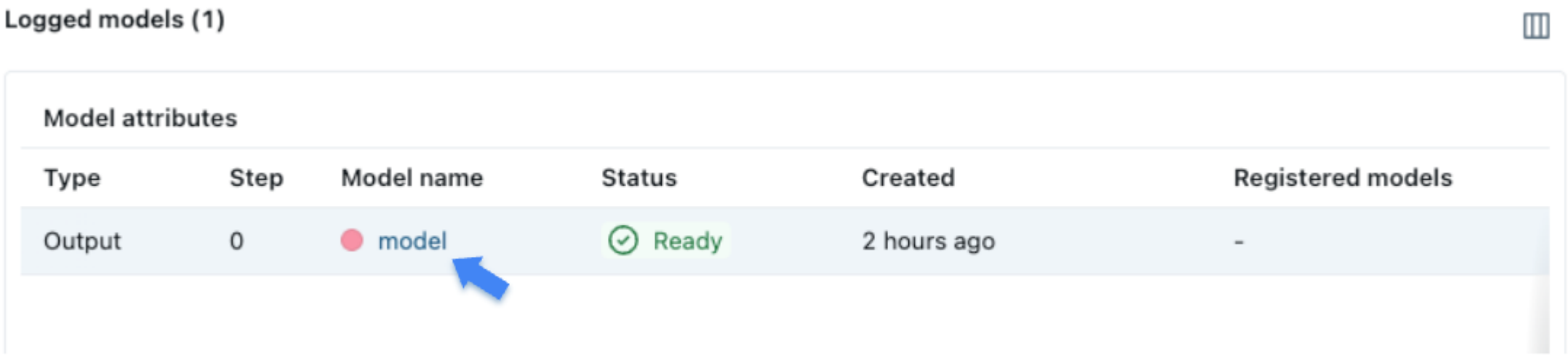
Step 4 - View the Run in the MLflow UI

- The Run detail page shows an overview of the run, its recorded metrics, hyper-parameters, tags, and more. Play around with the UI to see the different views and features.



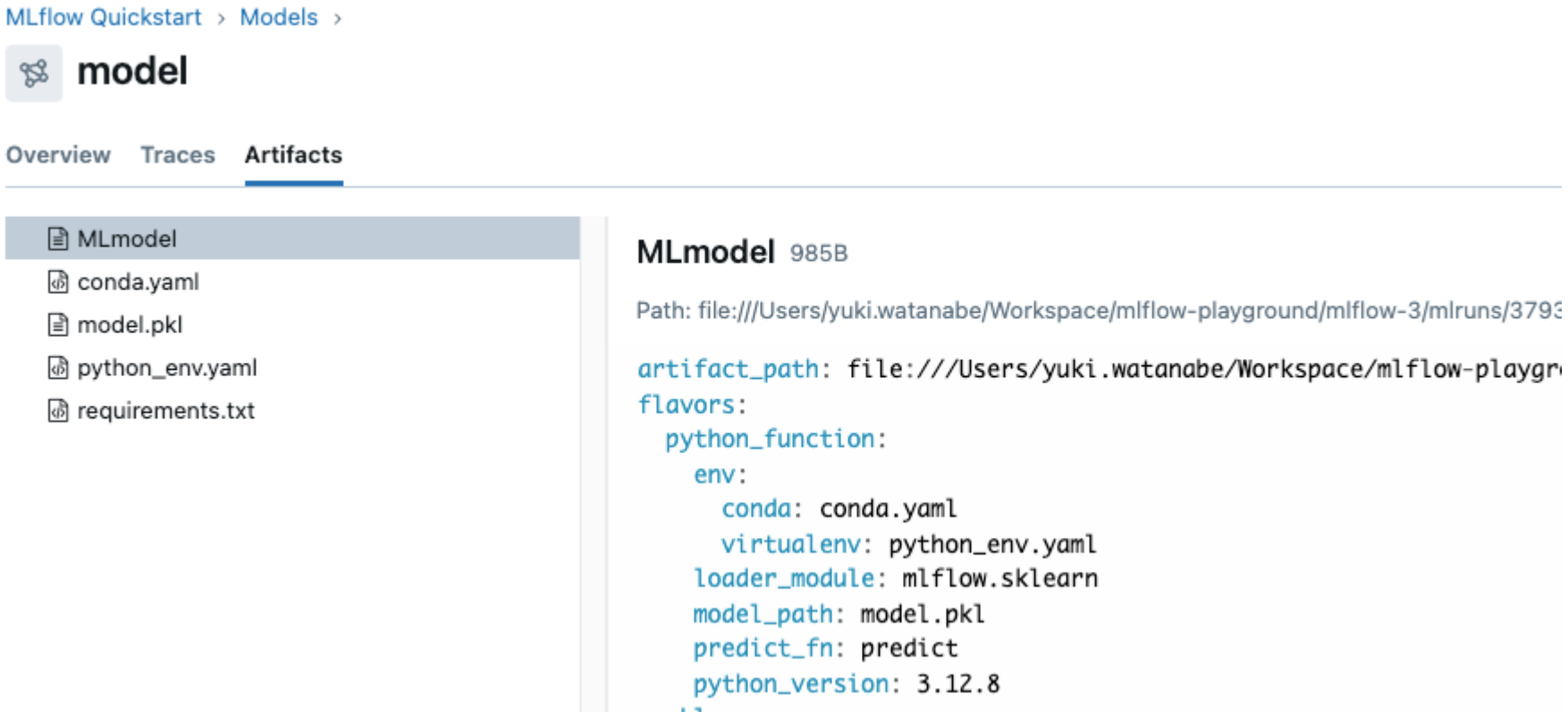
Step 4 - View the Run in the MLflow UI

- Scroll down to the "Model" section and you should see the model that was logged during training. Click on the model to view the details.



Step 4 - View the Run in the MLflow UI

- The model page displays similar metadata such as performance metrics and hyper-parameters.
- It also includes an "Artifacts" section that lists the files that were logged during training. You can also see environment information such as the Python version and dependencies, which are stored for reproducibility.



Step 5 - Log a model and metadata manually

- This is useful when you want to have more control over the logging process.
- **The steps that we will take are:**
  - Initiate an MLflow run context to start a new run that we will log the model and metadata to.
  - Train and test the model.
  - Log model parameters and performance metrics.
  - Tag the run for easy retrieval.

Step 5 - Log a model and metadata manually

```
# Start an MLflow run
with mlflow.start_run():
    # Log the hyperparameters
    mlflow.log_params(params)

    # Train the model
    lr = LogisticRegression(**params)
    lr.fit(X_train, y_train)

    # Log the model
    model_info = mlflow.sklearn.log_model(sk_model=lr, name="iris_model")

    # Predict on the test set, compute and log the loss metric
    y_pred = lr.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    mlflow.log_metric("accuracy", accuracy)

    # Optional: Set a tag that we can use to remind ourselves what this run was fo
    mlflow.set_tag("Training Info", "Basic LR model for iris data")
```

Step 6 - Load the model back for inference

- After logging the model, we can perform inference by:
  - Loading the model using MLflow's pyfunc flavor.
  - Running Predict on new data using the loaded model.
- **To load the model as a native scikit-learn model,** use mlflow.sklearn.load_model(model_info.model_uri) instead of the pyfunc flavor.

```
# Load the model back for predictions as a generic Python Function model
loaded_model = mlflow.pyfunc.load_model(model_info.model_uri)

predictions = loaded_model.predict(X_test)

iris_feature_names = datasets.load_iris().feature_names

result = pd.DataFrame(X_test, columns=iris_feature_names)
result["actual_class"] = y_test
result["predicted_class"] = predictions

result[:4]
```

The output of this code will look something like this:

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	actual_class	predicted_class
6.1	2.8	4.7	1.2	1	1
5.7	3.8	1.7	0.3	0	0
7.7	2.6	6.9	2.3	2	2
6.0	2.9	4.5	1.5	1	1

Thank you

- **Documentation:** https://mlflow.org/docs/latest/ml/
- Quick Start: https://mlflow.org/docs/latest/ml/getting-started/quickstart/
- Autologging: https://mlflow.org/docs/latest/ml/tracking/autolog/